

Degly Sebastián Pava Pava

✉ deglypavapava@gmail.com 🌐 https://deglypava.com 🎓 IMLEX and Systems and Computer Engineer

From: August 2021
To: September 2023
(Currently in Japan)

Erasmus Mundus Japan – Master of Science in Imaging and Light in Extended Reality (IMLEX)

- Quality Board Member and Communications Representative
- Triple Degree Program (Msc. Optics, Image, Computer Vision, Msc. Engineering, Msc. Computer Science)
- Toyohashi University of Technology (Japan), KU Leuven (Belgium), University of Eastern Finland (Finland), University Jean Monnet Saint-Etienne (France)

Tech: Photonics | Realtime 3D Rendering | Computational Imaging | CUDA | OpenMP | Computer Graphics | Physical Optics | Machine Learning | ARKit | ARCore | VR/AR.

WORK EXPERIENCE

Company Owner

Omni Applications L.L.C. (Florida, U.S.A) – Lucid Dreams Ltd. (Bogotá, Colombia)

Tech: Project Management | ThreeJS | iOS | XCode | Unreal | SparkAR | 360 | LensStudio | Blender | Jira | Slack

From: January 2018
To: August 2021

Lead XR Developer | Hackspace for Innovation and Visualization in Education

Faculty of Medicine, The University of British Columbia, Vancouver, B.C. (Canada)

- As part of THE HIVE Research Laboratory, contributed to The HoloBrain and MR Manikin / OCS projects by creating multiple 3D experiences and integrating them into a mixed reality environment using concepts of 3D visual computing. Utilized Unity 3D Engine to develop the experiences and released them through various devices, including Oculus Rift, HTC VIVE, and Extended Reality devices.

Tech: Unity | Microsoft HoloLens 1&2 | OpenXR | Mixed Reality Toolkit (MRTK) | Vuforia | Blender | Neuroanatomy | Azure | Cloud Computing | Dependency Injection | Reactive Programming

From: February 2017
To: September 2019

Lead Developer and Engineer at Pontifical Xavierian University

Department of Industrial and Systems Engineering at Pontifical Xavierian University, Bogotá (Colombia)

- Created a 3D Material Requirements Planning (MRP) simulation system using NetLogo to optimize production planning, scheduling, and inventory control. Currently used in production classes to train students in MRP and scheduling concepts.

Tech: NetLogo | Anylogic | Linear Algebra | Microsoft Office Suite | Adobe Suite | Research Techniques

EDUCATION AND TRAINING

Certifications:

VR / MR / XR Developer Meta Spark Creator AR Meta AR Developer AR & Video Streaming Services

Academic Recognitions:

Bachelor of Systems and Computer Engineering Master of Imaging and Light in Extended Realities

- Special Admission Program, best high school graduates of Colombia Scholarship at Universidad Nacional de Colombia
- IronHacks – Purdue University Fellowship
- Emerging Leaders of the Americas Program
- IMLEX Full Tuition Award
- ERASMUS Mundus Japan Scholarship
- COLFUTURO PCB Scholarship
- Manutech SLEIGHT Certificate Scholarship
- BRMI Academic Mobility Award

Trained in multiple research groups:

Visual Perception and Cognition Laboratory
Hackspace for Innovation and Visualization in Education
Research Center for Open Digital Innovation
Grupo de investigación en procesos químicos y bioquímicos

Toyohashi University of Technology - Japan
The University of British Columbia - Canada
Purdue University – United States
Universidad Nacional de Colombia

From August 2021
To: September 2023
From: January 2019
To: April 2021
From: June 2018
To: December 2018

- Engineered a VR research experience, leveraging eye tracking technology and drawing from human psychology and HCI principles to analyze pupil responses in an immersive virtual environment.
- Designed and developed advanced neuroanatomy applications using MRI (Magnetic Resonance Imaging) and VR (Virtual Reality) technologies for medical research and education.
- Contributed to organizing the Gold IronHacks Purdue-UNAL Hackathon 2018. Also served as a judge, assessing participants' websites for the best implementation of Open Data based on specific attributes.

Digital skills	Extended Realities	Programming	Optics / Physics	Research and Tools
	Unity Blender ThreeJS	C# Python C++	Photonics Colorimetry	ImageJ3D Healthcare IT
	AWS Android UX design	JavaScript NetLogo Metal	Eye-Tracking Robotics	TensorFlow Photon Realtime
	Vuforia XR SDKs Arduino	Git Swift WebXR	Spectral Imaging	Mendeley Adobe Suite

PERSONAL SKILLS

Spanish

Native proficiency

French

Limited Working Proficiency

English

Bilingual - IELTS (8.0) and iBT TOEFL (103 points)

Japanese

Furigana / Basic Speaking and Writing